

DIVIDEND DECISIONS

1. Dividend

It is return to shareholders i.e. Equity and Preference

2. Basic Terms

$$\text{Earning per share (EPS)} = \frac{\text{Earning Available for Equity}}{\text{Number of Equity Shares}}$$

$$\text{Dividend per share (DPS)} = \frac{\text{Amount of Dividend}}{\text{Number of Equity Shares}} = \text{EPS} \times \text{DP Ratio}$$

$$\text{Dividend Pay-out Ratio (DP Ratio)} = \frac{\text{DPS}}{\text{EPS}} \times 100 = 100 - \text{Retention Ratio}$$

$$\text{Retention Ratio (b)} = 100 - \text{DP Ratio}$$

$$\text{Dividend Rate} = \frac{\text{DPS}}{\text{Face Value}} \times 100$$

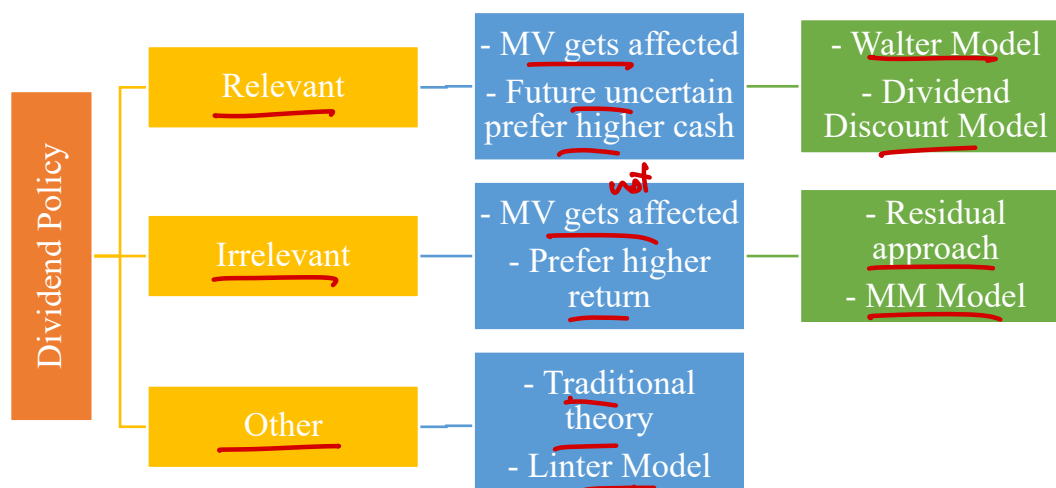
$$\text{Dividend Yield} = \frac{\text{DPS}}{\text{Market price per share}} \times 100$$

$$\text{Earning Yield} = \frac{\text{EPS}}{\text{Market price per share}} \times 100$$

$$\text{Price Earning Ratio (PE Ratio)} = \frac{\text{MPS}}{\text{EPS}}$$

$$\text{PE Ratio} = \frac{1}{K_e}$$

3. Dividend Policy



4. Walter Model

Assumptions of Walter Model

- The firm has infinite life
- All investments proposals of the firm are to be financed through retained earnings only
- The business risk complexion of the firm remains same. In other words, the rate of return on investment i.e. 'r' and the cost of capital of the firm i.e. 'Ke' are constant
- The firm operates in perfect capital market i.e. all investors are rational and information is freely available.
- There are no taxes
- There is no flotation or transaction costs

Explanation of Walter Model

As per this model, a company should or should not pay dividends depends upon whether it has got the suitable investment opportunities to invest the retained earnings or not.

The relationship between dividend and share price based on Walter's formula is shown below:

Market price of share $(P_0) = \frac{D + (E - D)(r/K_e)}{K_e}$

Where, E = Earning per share

D = Dividend per share

Ke = Cost of equity or rate of capitalization or discount rate

r = Rate of return on investment

<u>Position of r and Ke</u>	<u>Company</u>	<u>Optimum Dividend Payout Ratio</u>
$r > K_e$	Growth	Zero
$r < K_e$	Decline	100%
$r = K_e$	Constant	Every payout ratio is optimum

Question – 1

The following information is supplied to you:

	(₹)
Total Earnings	2,00,000
No. of equity shares (of ₹ 100 each)	20,000
Dividend paid	1,50,000
Price/Earnings ratio	12.50

Applying Walter's Model:

(i) Analyze whether the company is following an optimal dividend policy.

$$\text{Present } P_0 = 7.50 + (10 - 7.50) \left(\frac{0.10}{0.08} \right) = 132.81$$

$$\text{Best } P_0 = 0 + (10 - 0) \left(\frac{0.10}{0.08} \right) = 156.25$$

$$r = K_e \Rightarrow K_e = 10\% \Rightarrow P/E = \frac{1}{K_e} = \frac{1}{0.10} = 10 \text{ times}$$

(ii) Compute P/E ratio at which the dividend policy will have no effect on the value of the share.

(iii) Will you decide change if the P/E ratio is 8 instead of 12.5? Analyze.

$$\rightarrow K_e = \frac{1}{8} = 0.125 = 12.5\% \rightarrow 100\% \text{ DP Ratio}$$

Solution

$$EPS = \frac{\text{Total Earnings}}{\text{No. of equity shares}} = \frac{2,00,000}{20,000} = ₹ 10 \text{ per share}$$

$$DPS = \frac{\text{Dividend paid}}{\text{No. of equity shares}} = \frac{1,50,000}{20,000} = ₹ 7.50 \text{ per share}$$

$$\text{Rate of return, } r = \frac{\text{Total earnings}}{\text{Total equity value}} \times 100 = \frac{2,00,000}{20,000 \times 100} \times 100 = 10\%$$

$$K_e = \frac{1}{P/E \text{ Ratio}} = \frac{1}{12.50} = 0.08 = 8\%$$

$$P_0 = \frac{10 + \frac{(10-10)(\frac{0.10}{0.125})}{0.125}}{0.125} = 80$$

(i) At present, company pays dividend of ₹ 7.50 per share at which market price comes at:

$$P_0 = \frac{D + (E-D)(r+K_e)}{K_e} = \frac{7.50 + (10-7.50)(0.10+0.08)}{0.08} = ₹ 132.81$$

Since, $r(10\%)$ is greater than $K_e(8\%)$, thus as per Walter Model, the optimum dividend payout ratio should be zero. Market price at zero dividend is:

$$P_0 = \frac{D + (E-D)(r+K_e)}{K_e} = \frac{0 + (10-0)(0.10+0.08)}{0.08} = ₹ 156.25$$

Thus, theoretically the market price of the share can be increased by adopting a zero payout.

(ii) As per Walter Model, when $r = K_e$, then dividend policy will have no effect on the value of the share price. Thus, $K_e = 10\%$ will be the required level.

$$P/E \text{ Ratio} = \frac{1}{K_e} = \frac{1}{0.10} = 10$$

(iii) If P/E ratio is 8, then $K_e = \frac{1}{P/E \text{ Ratio}} = \frac{1}{8} = 0.125 = 12.5\%$

Now $r(10\%)$ is less than $K_e(12.50\%)$. As per Walter model, in such case it is advisable to distribute maximum dividend to maximize the value of the share.

If 100% dividend is given then, price of share is:

$$P_0 = \frac{D + (E-D)(r+K_e)}{K_e} = \frac{10 + (10-10)(0.10+0.125)}{0.125} = ₹ 80$$

$$EPS = \frac{50 - (200 \times 13\%)}{6} = 4$$

Question – 2

The following figures are extracted from the annual report of RJ Ltd.:

Net Profit	→ ₹ 50 Lakhs
Outstanding 13% preference shares	→ ₹ 200 Lakhs
No. of Equity shares	→ 6 Lakhs
Return on Investment (r)	→ 25%
Cost of Capital (K _e)	→ 15%

$$40 = \frac{D + (4-D)(\frac{0.25}{0.15})}{0.15}$$

$$6 = \frac{D + 6.67 - (D)(1.67)}{0.15}$$

$$0.67 = (0.67)(D)$$

$$D = 1$$

You are required to compute the approximate dividend pay-out ratio by keeping the share price at ₹ 40 by using Walter's Model.

$$\rightarrow \frac{1}{4} \times 100 = 25\%$$

Solution

Earning available for equity = Net Profit – Preference Dividend
 = 50 lakhs – (200 lakhs × 13%) = ₹ 24 Lakhs

$$\text{Earnings per share} = \frac{\text{Earning available for Equity}}{\text{No. of Equity Shares}} = \frac{24,00,000}{6,00,000} = ₹ 4$$

As per Walter Model, $P = \frac{D + (E - D)(r + Ke)}{Ke}$

Where,

P = Market price per share = ₹ 40

E = Earnings per share = ₹ 4

D = Dividend per share

r = Return earned on investment = 25% = 0.25

Ke = Cost of equity capital = 15% = 0.15

$$\therefore P = \frac{D + (4 - D)(0.25 + 0.15)}{0.15}$$

$$40 = \frac{D + (4 - D)(1.6667)}{0.15}$$

$$6 = D + 6.667 - (1.667)D$$

$$0.667D = 0.6667$$

$$D = ₹ 1$$

$$\text{Required dividend pay-out ratio} = \frac{\text{Dividend per share}}{\text{Earning per share}} \times 100 = \frac{1}{4} \times 100 = 25\%$$

Question – 3

Profitable FMCG company PQR Ltd. is examining its dividend policy considering potential future investment opportunities. The company's objective is to maintain a good financial strategy by balancing market valuation, financing requirements, and shareholder expectations.

$$\text{EPS} = \frac{102}{20} = 5$$

$$A \rightarrow P_0 = \frac{5 + (5 - 5) \left(\frac{0.15}{0.12} \right)}{0.12} = 41.67$$

Following information has been provided:

Equity Share Capital: ₹20,00,000 (₹10 per share fully paid)

Reserves and Surplus: ₹30,00,000

Earnings After Tax (EAT): ₹10,00,000

Return on Investment (r): 15%

Cost of Equity (Ke): 12%

Market Price per Share: ₹80

Number of Equity Shares: 2,00,000

Current Debt-Equity Ratio: 0.5:1

Planned Investment Requirement: ₹12,00,000 (next financial year)

$$\text{Current Debt} = 502 \times 0.5 = 251$$

$$B \rightarrow P_0 = \frac{2.50 + (5 - 2.50) \left(\frac{0.15}{0.12} \right)}{0.12} = 46.88$$

$$C \rightarrow P_0 = \frac{0 + (5 - 0) \left(\frac{0.15}{0.12} \right)}{0.12} = 52.08$$

A	RS	Debt	Equity	Ratio	
-	122	→	$\frac{251 + 122}{502} = 0.74:1$	⊗	
B	52	72	→	$\frac{251 + 72}{502 + 52} = \frac{323}{554} = 0.58:1$	⊗
C	102	22	→	$\frac{251 + 22}{502 + 102} = 0.45:1$	⊗

The Board of Directors of the company is considering three dividend payout options. Option A – 100% payout; Option B – 50% payout; and Option C – 0% payout.

$$5 \times 50\% = 2.50$$

Further, because of the market conditions, no fresh equity will be raised. Any financial shortage must be covered by debt, but not more than a 1:1 debt-to-equity ratio. The corporate tax rate is 30%. Interest on debt is 10% per annum.

You are required to:

- Calculate the market price per share under each dividend payout option using Walter's Model.
- Evaluate the feasibility of internally funding the ₹12,00,000 investment under each payout scenario.
- Determine the revised Debt-Equity Ratio assuming the shortfall is met through borrowing if the company adopts a 50% payout.
- Recommend the most suitable dividend policy.

Solution

(i) Market Price Per Share using Walter's Model

Walter's Model Formula:

$$P = \frac{[D + (r / K_e) \times (E - D)]}{K_e}$$

Option A: 100% payout (D = ₹ 5)

$$P = \frac{[5 + (15\% / 12\%) \times (5 - 5)]}{0.12} = ₹ 41.67$$

Option B: 50% payout (D = ₹ 2.5)

$$P = \frac{[2.5 + (15\% / 12\%) \times (5 - 2.5)]}{0.12} = ₹ 72.92$$

Option C: 0% payout (D = ₹ 0)

$$P = \frac{[0 + (15\% / 12\%) \times 5]}{0.12} = ₹ 104.17$$

Since the company earns more on retained earnings ($r > K_e$), lower payout leads to higher market price per share.

(ii) Feasibility of Funding ₹ 12,00,000 Investment Internally

Investment Requirement = ₹ 12,00,000

EAT = ₹ 10,00,000

Retained earnings depend on payout ratio.

The present situation can be summarized in the following table:

Option	Retained Earnings (₹)	Shortfall (₹)	Feasibility
100% payout	0	12,00,000	Feasible
50% payout	5,00,000	7,00,000	Feasible
0% payout	10,00,000	2,00,000	Feasible

(iii) **Revised Debt-Equity Ratio (if 50% payout is chosen)**

Existing Equity = ₹ 50,00,000

Existing Debt = $0.5 \times ₹ 50,00,000 = ₹ 25,00,000$

New Equity (post-retention): ₹ 55,00,000

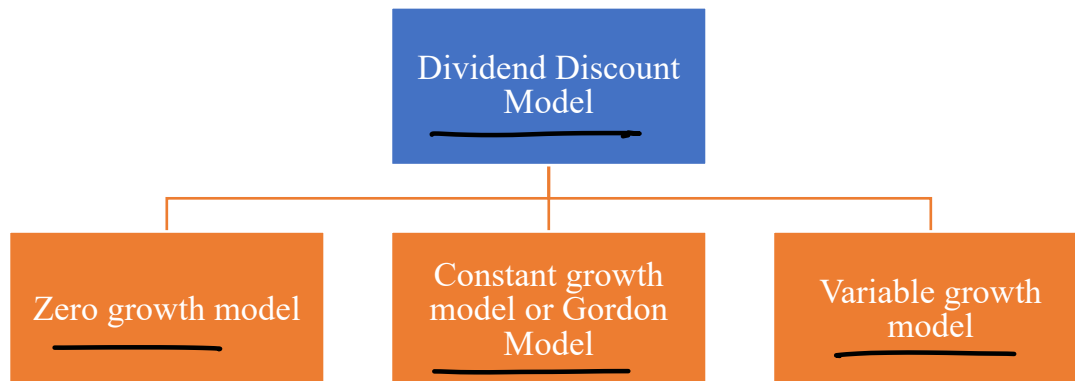
New Debt (after ₹ 7,00,000 borrowing): ₹ 32,00,000

Revised debt-equity Ratio = $₹ 32,00,000 / ₹ 55,00,000 = 0.582$

It is well within the 1:1 cap.

(iv) **Recommendation**

0% payout is the best option as it has the highest market price which increases shareholders wealth.

5. Dividend Discount Model**6. Zero Growth Model**

$$\text{Share's Intrinsic Value (P}_0\text{)} = \frac{\text{Annual Dividend}}{\text{Require Rate of Return}} = \frac{D}{K_e}$$

7. Constant Growth Model or Gordon Model**Assumptions of Gordon Model**

- The firm is an all equity firm i.e. it has no debt
- All investment proposals of the firm are to be financed through retained earnings only and no external finance is available to the company.
- The business risk complexion of the firm remains same. In other words, rate of return on investment i.e. 'r' and cost of capital of the company i.e. 'Ke' are constant.
- The firm has an infinite life
- The cost of capital besides being constant is more than the growth rate i.e. $K_e > g$
- The growth rate of the company 'g' is the product of its retention ratio 'b' and its rate of return 'r' i.e. $g = br$

Explanation of model

This model suggests that, investors prefer what is available at present compared to what may be available in future. Investors are certain of receiving incomes from dividends than from future capital gains. Investors will prefer to pay a higher price for shares on which current dividends are paid as compared to those shares on which future capital gains are expected.

Under this model, market price of share can be calculated as follows:

$$\text{Market Price (P}_0\text{)} = \frac{D_1}{K_e - g} = \frac{D_0(1+K_e)}{K_e - g}$$

Where, D_1 = Next expected dividends or dividends payable at the end of the year

D_0 = current year Dividend

K_e = cost of Equity capital or expected rate of return

G = growth rate of dividends = br

$$\text{EPS} = \frac{30 - (100 \times 12\%)}{30} = 6$$

$$\text{(i) } P_0 = \frac{1.5}{0.16 - (0.75)(0.20)} = 150$$

$$\text{(ii) } P_0 = \frac{3}{0.16 - (0.50)(0.20)} = 50$$

Question – 4

The following figures are collected from the annual report of XYZ Ltd.:

Net Profit	→ ₹ 30 lakhs
Outstanding 12% Preference shares	→ ₹ 100 lakhs
No. of equity shares	→ 3 lakhs
Return on Investment (r)	→ ✓ 20%
Cost of capital i.e. (K_e)	→ 16%

$$\text{(iii) } P_0 = \frac{6}{0.16 - (0)(0.20)} = 37.5$$

Calculate price per share using Gordon's Model when dividend pay-out is (i) 25%; (ii) 50% and (iii) 100%.

6

Solution

As per Gordon's Formula, $P = \frac{E(1-b)}{K_e - (b \times r)}$

$$\text{Earning per share (E)} = \frac{\text{Earning available for equity}}{\text{No. of equity shares}} = \frac{30,00,000 - 12,00,000}{3,00,000} = ₹ 6$$

$$\text{Cost of equity (K}_e\text{)} = 16\% = 0.16$$

$$\text{Return on investment (r)} = 20\% = 0.20$$

$$\text{(i) When dividend pay-out ratio is 25\%, } P = \frac{6 \times 0.25}{0.16 - (0.75 \times 0.20)} = \frac{1.50}{0.16 - 0.15} = ₹ 150$$

$$\text{(ii) When dividend pay-out ratio is 50\%, } P = \frac{6 \times 0.50}{0.16 - (0.50 \times 0.20)} = \frac{3}{0.16 - 0.10} = ₹ 50$$

$$\text{(iii) When dividend pay-out ratio is 100\%, } P = \frac{6 \times 1}{0.16 - (0 \times 0.20)} = \frac{6}{0.16} = ₹ 37.50$$

$$P_0 = \frac{10(1+0.054)}{0.0885 - 0.054} = \frac{10.54}{0.0345} = 305.51$$

Question – 5

HM Ltd. is listed on Bombay Stock Exchange which is currently been evaluated by Mr. A on certain parameters. Mr. A collated following information:

- (a) The company generally gives a quarterly interim dividend ₹ 2.5 per share in the last dividend declared. $D_0 = 2.5 \times 4 = ₹ 10$
- (b) The company's sales are growing by 20% on a 5-year Compounded Annual Growth Rate (CAGR) basis, however the company expects following retention amounts against probabilities mentioned as contention is dependent upon cash requirements for the company. Rate of return is 10% generated by the company. $g = (b)(r) = (0.54)(0.10) = 0.054$
- (c) The current risk-free rate is 3.75% and with a beta of 1.2 company is having a risk premium of 4.25%. $b = (0.30)(50) + (0.40)(60) + (0.30)(50) = 54\%$

Situation	Prob.	Retention Ratio
A	30%	50%
B	40%	60%
C	30%	50%

- (c) The current risk-free rate is 3.75% and with a beta of 1.2 company is having a risk premium of 4.25%. $K_e = 3.75 + (4.25)(1.2) = 8.85\%$

You are required to help Mr. A in calculating the current market price using Gordon's formula.

Solution

$$D_0 = 2.5 \times 4 = ₹ 10 \text{ (annually)}$$

$$\text{Expected retention ratio} = b = (50 \times 0.30) + (60 \times 0.40) + (50 \times 0.30) = 54\%$$

$$g = (b)(r) = (0.54)(0.10) = 0.054 = 5.4\%$$

$$K_e = R_f + (\text{Beta} \times \text{Risk Premium}) = 3.75 + (1.2 \times 4.25) = 8.85\%$$

$$P_0 = \frac{D_0(1+g)}{K_e - g} = \frac{10(1+0.054)}{0.085 - 0.054} = \frac{10.54}{0.0345} = ₹ 305.51$$

$$\begin{aligned} \text{MPS} &= \text{EPS} \times \text{P/E} \\ \frac{D_1}{K_e - g} &= 10 \times 12 \\ \frac{10(1-b)}{0.20 - (b)(0.25)} &= 120 \\ 10 - 10b &= 24 - 30b \\ 20b &= 14 \Rightarrow b = \frac{14}{20} = 0.70 \end{aligned}$$

Question – 6

Y Ltd. produces energy drinks in different flavours. Due to high demand of its product, the rate of return on its earnings is 25%. Currently, the company retains 60% of its earnings and distributes the rest. The current P/E ratio is 8 and earnings per share is ₹10.

According to Gordon's Model:

- (i) What will be retention ratio if the company wants to maintain its P/E ratio to 12 in current year, given that the expected rate of return for an investor is 20%?
- (ii) What will be the expected price per share after one year if Y Ltd. achieves above-mentioned targeted P/E ratio? $g = 0.70 \times 0.25 = 0.175$ $\text{Price} = 120 \times (1+0.175) = 141$
- (iii) Will there be any change in retention ratio if the company wants to maintain its P/E ratio to 10 in current year, given that the expected rate of return for an investor is 17.50%?

Solution

The question can be solved with the help of Justified P/E Ratio linked to Gordon Growth Model.

(i) Calculation of Retention Ratio

$$\text{PE Ratio} = \frac{\text{MPS}}{\text{EPS}}$$

$$\begin{aligned} \text{MPS} &= \frac{D}{K_e - g} \\ 10 \times 10 &= \frac{10(1-b)}{0.175 - (b)(0.25)} \\ 17.5 - 25b &= 10 - 10b \\ 15b &= 7.5 \\ b &= \frac{7.5}{15} = 0.50 \end{aligned}$$

$$12 = \frac{MPS}{10}$$

$$120 = MPS$$

$$120 = \frac{E(1-b)}{K_e - (b)(r)}$$

$$120 = \frac{10(1-b)}{0.20 - (b)(0.25)}$$

$$12 = \frac{(1-b)}{0.20 - (b)(0.25)}$$

$$2.4 - 3b = 1 - b$$

$$1.4 = 2b$$

$$b = 0.70 \text{ or } 70\%$$

So, Retention Ratio is 70%, if the company wants to maintain its P/E ratio to 12 in the current year.

(ii) Calculation of expected price of share

$$MPS = EPS \times PE \text{ Ratio} = 12 \times 10 = ₹120$$

$$\text{Growth rate} = b \times r = 0.70 \times 0.25 = 0.175 \text{ or } 17.5\%$$

$$\text{Expected price after one year} = ₹120 \times (1 + 0.175) = ₹141$$

Therefore, the expected price per share after one year is ₹141 if Y Ltd. achieves the targeted PE ratio of 12.

(iii) Calculation of Change in Retention Ratio

$$PE \text{ Ratio} = \frac{MPS}{EPS}$$

$$10 = \frac{MPS}{10}$$

$$100 = MPS$$

$$100 = \frac{E(1-b)}{K_e - (b)(r)}$$

$$100 = \frac{10(1-b)}{0.175 - (b)(0.25)}$$

$$10 = \frac{(1-b)}{0.175 - (b)(0.25)}$$

$$1.75 - 2.5b = 1 - b$$

$$0.75 = 1.5b$$

$$b = 0.50 \text{ or } 50\%$$

So, Retention Ratio is 50%, if the company wants to maintain its P/E ratio to 10 in the current year.

$$ROE = \frac{\text{EAFE}}{\text{Eq. Sh. fund}} \times BVPS = \frac{\text{Eq. Sh. fund}}{\text{No. of Eq. Sh.}} \Rightarrow \frac{\text{EAFE}}{\text{No. of Eq. Sh.}} = EPS = 125 \times 15\% = 18.75$$

Question – 7	$b = 65\%$ $DP = 35\%$ $D = 18.75 \times 35\% = 6.5625$ $g = (0.15)(0.65) = 9.75\%$
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Return on Equity (ROE) is Satva Limited is 15% and the capitalization rate applicable to the company is at 20%. Satva Limited's Book Value per share (BVPS) is ₹125. Calculate the intrinsic value of the share today using Gordon's model and Walter's model if the company's policy is to retain 65% of the earning.

$$\frac{6.5625(1 + 0.0975)}{0.20 - 0.0975} = 70.27$$

$$\frac{6.5625 + (18.75 - 6.5625)\left(\frac{0.15}{0.20}\right)}{0.20} = 78.52$$

Solution

$$\text{EPS} = \text{ROE} \times \text{BVPS (WN 1)}$$

$$\text{EPS} = 0.15 \times 125 = ₹ 18.75$$

$$\text{Growth} = \text{ROE} \times \text{Retention Ratio}$$

$$= 0.15 \times 0.65$$

$$= 9.75\%$$

$$D_1 = D_0 (1 + g)$$

$$= (18.75 \times 35\%)(1 + 0.0975)$$

$$= ₹ 7.20$$

Intrinsic Value of share today - Gordon's Formula

$$P_0 = \frac{D_1}{K_e - g}$$

$$= \frac{7.20}{0.20 - 0.0975}$$

$$P_0 = ₹ 70.24$$

Intrinsic Value of share today - Walter's Model

$$P_0 = \frac{D + \frac{r}{K_e}(E - D)}{K_e}$$

Here D = D₀ assuming it would remain constant through infinity

$$P_0 = \frac{6.5625 + \frac{0.15}{0.20}(18.75 - 6.5625)}{0.20}$$

$$P_0 = ₹ 78.51$$

WN 1 - Relationship between ROE-EPS-BVPS

$$\text{ROE} = \frac{\text{Earnings for Equity Shareholders}}{\text{Equity shareholders funds}}$$

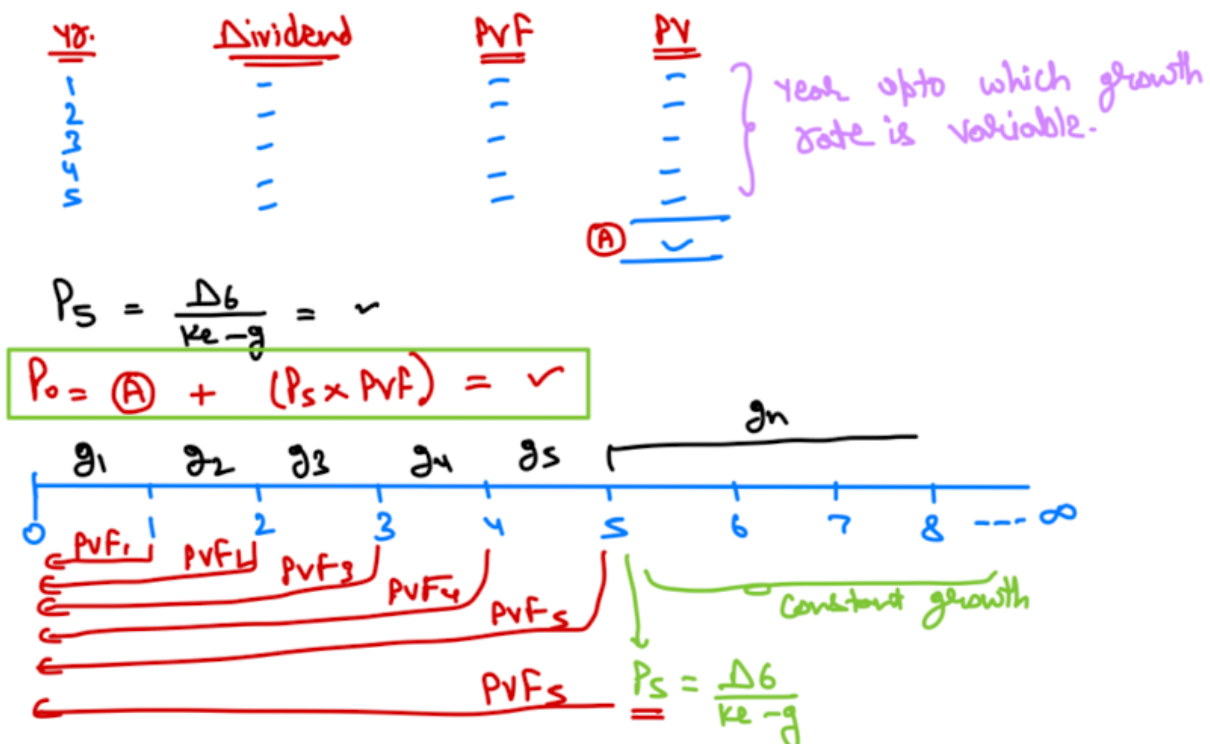
If we divide the numerator and denominator with "No of equity shares"

$$\text{ROE} = \frac{\text{Earnings for Equity Shareholders} / \text{No of equity shares}}{\text{Equity shareholders funds} / \text{No of equity shares}}$$

Therefore, ROE = EPS / BVPS

8. Variable Growth Rate Model

Price = Present value of all future benefits (i.e. dividend) from shares



Question – 8

S Ltd. is foreseeing a growth rate of 12% p.a. in the next two years. The growth rate is likely to be 10% for the third and fourth year. After that the growth rate is expected to stabilize at 8% p.a. If the last dividend was ₹ 1.50 per share and the investor's required rate of return is 16%, determine the current value of equity share of the company. The present value factors at 16% are as below:

Year	1	2	3	4
PVF	0.862	0.743	0.641	0.552

Solution

Year	Particulars	Amount	PVF @ 16%	Present Value
1	Dividend	$1.50 \times (1+0.12) = 1.68$	0.862	1.45
2	Dividend	$1.68 \times (1+0.12) = 1.88$	0.743	1.40
3	Dividend	$1.88 \times (1+0.10) = 2.07$	0.641	1.33
4	Dividend	$2.07 \times (1+0.10) = 2.28$	0.552	1.24
	Total			5.42

Price at end of 4th year, $P_4 = \frac{D_5}{K_e - g} = \frac{2.28(1+0.08)}{0.16-0.08} = ₹ 30.78$

Current value of equity share = ₹ 5.42 + (₹ 30.78 × 0.552) = ₹ 22.41

CP = ₹. 23 → over value (Not to Buy)
 CP = ₹. 21 → under-value (Yes to Buy)

9. ^(IV) Intrinsic Value Vs ^(CV) Current Value

IF IV > CV – Undervalued security or recommend to buy

IF IV < CV – Overvalued security or not recommend to buy

10. Modigliani And Miller (MM) Model

Assumptions of MM Hypothesis

- The capital markets operate in perfect market conditions
- Freely information is available to all investors
- Securities are infinitely divisible
- There are no flotation costs
- There are not taxes i.e. neither on dividend nor on capital gain
- Investors are rational and well informed about the risk and return of all the securities
- The company has a fixed investment policy
- Cost of equity remains constant

Explanation of MM Hypothesis

- According to MM theory, earnings are the sole criteria on which market value of the firm is determined. Market value is not influenced by the split of earnings between dividends and retained earnings.
- The underlying intuition for this theory is simple – Company that pay more dividends offer less price appreciation but provide the same total return to shareholders, given the risk characteristics of the company. The investors should be indifferent of receiving their returns in the form of current dividends or in the form of price increase in the market.
- This can be computed with the help of following formula:

$$P_0 = \frac{P_1 + D_1}{1 + K_e}$$

Where, P_0 = Market price of share today

P_1 = Market price of share at the end of year 1

D_1 = Dividend per share at the end of year 1

K_e = Cost of equity share capital or discount rate or rate of capitalization

Additional number of shares to be issued at the end of year 1 = $\Delta n = \frac{I_1 - (E - D)}{P_1}$

Where, I_1 = Amount required for investment

E = Total earnings of the company

D = Total dividend to be distributed

P_1 = Market price of share at the end of year 1

Current Value of Firm = No. of equity shares $\times$ MPS = $\frac{(n + \Delta n)(P_1) - I + E}{(1 + K_e)}$

Question – 9

$$n = \frac{1000}{100} = 10$$

ZX Ltd. has a paid-up share capital of ₹ 1,00,00,000 face value for ₹ 100 each. The current market price of the shares is ₹ 100 each. The Board of Directors of the company has an agenda of meeting to pay a dividend of 50% to its shareholders. The company expects a net income of ₹ 75,00,000 at the end of the current financial year. Company also plans for a capital expenditure for the next financial year for a cost of ₹ 95,00,000, which can be financed through retained earnings and issue of new equity shares. Company's desired rate of investment is 15%.

Required:

Following the Modigliani-Miller (MM) Hypothesis, determine value of the company when:

- It does not pay dividend and
- It does pay dividend

Solution

Working Notes:

- Market price per share = ₹ 100

As per MM Model, the price of the share (if dividend is paid):

$$P_0 = \frac{(D_1 + P_1)}{(1 + K_e)} \Rightarrow 100 = \frac{50 + P_1}{(1 + 0.15)} \Rightarrow P_1 = ₹ 65$$

- As per MM Model, the price of the share (if dividend is not paid):

$$P_0 = \frac{(D_1 + P_1)}{(1 + K_e)} \Rightarrow 100 = \frac{0 + P_1}{(1 + 0.15)} \Rightarrow P_1 = ₹ 115$$

The number of new equity shares can be found as follows:

	Dividends Distributed (₹)	Dividends Not Distributed (₹)
Net Income	75,00,000	75,00,000
Total Dividends (1,00,000 × 50)	50,00,000	-
Retained Earnings	25,00,000	75,00,000
Investment Budget	95,00,000	95,00,000
Amount to be raised by new issues	70,00,000	20,00,000
Relevant market Price (₹ Per share)	65	115
No. of shares to be issued	1,07,692	17,391

- Value of ZX Ltd. when dividends are not paid

$$\text{Value} = \frac{(n + \Delta n)P_1 - I + E}{1 + K_e} = \frac{[(1,00,000 + 17,391) \times 115] - 95,00,000 + 75,00,000}{(1 + 0.15)} = ₹ 1,00,00,000$$

- Value of ZX Ltd. when dividends are paid

$$\text{Value} = \frac{(n + \Delta n)P_1 - I + E}{1 + K_e} = \frac{[(1,00,000 + 1,07,692) \times 65] - 95,00,000 + 75,00,000}{(1 + 0.15)} = ₹ 1,00,00,000$$

Question – 10

Ordinary shares of a listed company are currently trading at ₹ 10 per share with two lakh shares outstanding. The company anticipates that its earnings for next year will be ₹ 5,00,000. Existing cost of capital for equity shares is 15%. The company has certain investment proposals under discussion which will cause an additional 26,089 ordinary shares to be issued if no dividend is paid or an additional 47,619 ordinary shares to be issued if dividend is paid.

Applying the MM hypothesis on dividend decisions, calculate the amount of investment and dividend that is under consideration by the company.

Solution**When dividend is not paid**

$$P_0 = \frac{P_1 + D_1}{(1 + K_e)}$$

$$10 = \frac{P_1 + 0}{(1 + 0.15)}$$

$$P_1 = ₹ 11.50$$

$$\text{Number of shares} = \frac{I - E + nD_1}{P_1}$$

$$26,089 = \frac{I - 5,00,000 + 0}{11.50}$$

$$I = 8,00,024$$

When dividend is paid

$$P_0 = \frac{P_1 + D_1}{(1 + K_e)}$$

$$10 = \frac{P_1 + D_1}{(1 + 0.15)}$$

$$P_1 + D_1 = 11.50$$

$$P_1 = 11.50 - D_1 \dots\dots\dots(1)$$

$$\text{Number of shares} = \frac{I - E + nD_1}{P_1}$$

$$47,619 = \frac{8,00,024 - 5,00,000 + (2,00,000)(D_1)}{P_1}$$

$$47,619(P_1) = 3,00,024 + (2,00,000)(D_1)$$

$$47,619(11.5 - D_1) = 3,00,024 + (2,00,000)(D_1)$$

$$5,47,618.50 - 47,619(D_1) = 3,00,024 + (2,00,000)(D_1)$$

$$2,47,619(D_1) = 2,47,594.5$$

$$D_1 = ₹ 1$$

11. Traditional Model

According to the traditional model, the stock market places considerably more weight on dividends than on retained earnings. For them, the stock market is overwhelmingly in favour of liberal dividends as against niggardly dividends. Their view is expressed quantitatively in the following valuation model:

$$\text{Market Price (P)} = m \left(D + \frac{E}{3} \right)$$

Where, P = Market Price per share

D = Dividend per share

E = Earnings per share

M = Multiplier

$$P = 9 \left(D + \frac{E}{3} \right)$$

$$P = 9 \left((E \times 0.4) + \frac{E}{3} \right)$$

$$P = \frac{3}{1} \left[\frac{(1.2)E + E}{2} \right]$$

$$P = 3 (2.2) (E)$$

$$\frac{P}{E} = 6.6$$

$$\frac{P}{E} \text{ Ratio} = 6.6 \checkmark$$

Question – 11

The dividend payout ratio of S Ltd. is 40%. If the company follows traditional approach to dividend policy with a multiplier of 9, what will be the MPS/EPS or PE Ratio of S Ltd.

Solution

As per Traditional approach, $P_0 = m \times (D + E/3)$

$m = \text{multiplier} = 9$

$D = \text{Dividend per share} = E \times 40\% = 0.40 \times E$

$P_0 = \text{Market price}$

$$P_0 = m \times (D + E/3)$$

$$P_0 = 9 \times \left[(E \times 0.40) + \frac{E}{3} \right]$$

$$P_0 = 9 \times \frac{(1.20E + E)}{3}$$

$$P_0 = 3 \times 2.20 \times E$$

$$\frac{P_0}{E} = 6.60$$

$$\frac{MPS}{EPS} = 6.60$$

$$\text{PE Ratio} = 6.60$$

12. Linter Model

This model states that, the current year's dividend is dependent on current year's earnings and last year's dividends. Similarly, last year dividends' is based on last year's earnings and the second previous year's dividends. If the circumstances appeared to warrant a large increase in their company's dividend's then the firm would move only steadily towards their target payment. Conservative firms move slowly towards its target and hence would have a low adjustment rate.

$$D_1 = D_0 + [(EPS \times \text{Target payout}) - D_0] \times Af$$

Where, D_1 = Dividend in year 1 or next expected dividend

D_0 = Last year dividend

EPS = Earnings per share

Af = Adjustment factor

$$14 = 12 + [(E \times 0.4) - 12] (0.6)$$

$$2 = (E)(0.24) - 7.2$$

$$E(0.24) = 9.2$$

$$E = \frac{9.2}{0.24} = 38.3333$$

$$MPS = E \times PE \\ = 38.3333 \times 9 = 344.97$$

Question – 12

The target payout ratio for S Ltd. is 0.4. The dividend per share for the current year is ₹ 14. The dividend per share in previous year was ₹ 12. The weightage given to the current year earnings is 0.60. The number of equity shares outstanding in the company is 10,00,000. If the P/E multiple is 9, applying Lintner Model of dividend policy to the company, compute the market capitalization of the company.

$$10 \times 344.97 = 3449.7$$

Solution

As per Linter Model, $D_1 = D_0 + [(E \times \text{Target Payout}) - D_0] \times AF$

D_1 = Next expected dividend = 14

D_0 = Last paid dividend = 12

Target payout = 0.4

AF = Adjustment Factor = 0.6

$$\therefore 14 = 12 + [(E \times 0.40) - 12] \times 0.60$$

$$2 = (0.4E - 12) \times 0.60$$

$$3.33 = 0.4E - 12$$

$$0.4E = 15.33$$

$$E = 38.33$$

$$\therefore \text{Earning per share} = E = ₹ 38.33$$

$$PE \text{ Ratio} = MPS \div EPS$$

$$MPS = PE \text{ Ratio} \times EPS = 9 \times 38.33 = ₹ 344.97$$

$$\text{Market capitalization} = \text{No. of equity shares} \times MPS = 10,00,000 \times 344.97 = ₹ 3,449.7 \text{ lakhs}$$

$$EPS = \frac{270}{18} = 15$$

$$\text{No. of Eq. Sh. before BB} = \frac{180}{15} = 12,000$$

$$\text{No. of Eq. Sh. after Buyback} = 12,000 - 2,000 = 10,000$$

Question – 13

Paarath Limited had recently repurchased 20,000 equity shares at a premium of 10% to its prevailing market price. The book value per share (after repurchasing) is ₹ 193.20.

Other Details of the company are as follows:

Earnings of the company (before buyback) = ₹ 18,00,000

Current MPS is ₹ 270 with a P/E Ratio of 18.

Calculate the Book Value per share of the company before the re-purchase.

$$\begin{aligned} \text{BV after buyback} &= 10,000 \times 193.20 = 1,93,20,000 \\ (+) \text{Buyback } (20,000 \times 297) &= 59,40,000 \\ \text{BV before buyback} &= 2,52,60,000 \\ \text{No. of Sh.} &= 12,000 \\ \text{BV before buy} &= 210.5 \end{aligned}$$

Solution

$$\begin{aligned} \text{(i) No. of equity shares (before buyback)} &= \frac{\text{Total Earnings (before buyback)}}{EPS} \\ &= \frac{18,00,000}{(270 \div 18)} = 12,000 \text{ shares} \end{aligned}$$

$$\text{(ii) Buyback price} = ₹ 270 + 10\% \text{ Premium} = ₹ 297$$

$$\text{(iii) No. of equity shares (after buyback)} = 12,000 - 2,000 = 10,000 \text{ shares}$$

$$\text{(iv) Total book value of equity (after buyback)} = 10,000 \times 193.20 = ₹ 1,93,20,000$$

$$\begin{aligned} \text{Now, Total BV of equity (after buyback)} &= \text{Total BV (before buyback)} - \text{Amount of buyback} \\ 1,93,20,000 &= \text{Total BV (before buyback)} - (20,000 \times 297) \end{aligned}$$

$$\text{Total BV (before buyback)} = ₹ 2,52,60,000$$

$$\text{BV per share (before buyback)} = 2,52,60,000 \div 12,000 = ₹ 210.50 \text{ per share}$$